johnston-linear-matrix-algebra: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_TAB_001	89	—	`\begin{tabular}[t]{|l|l|l|} \hline \multicolumn{3}{|l|}{Alice's lists:} \\ \hline Bob & Cora & Devon \\ \hline Alice & Alice & Alice \\ \hline Devon & & Bob \\ \hline \end{tabular}`	Alice's lists: Bob | Cora | Devon Alice | Alice | Alice Devon | | Bob	Alice's lists: **Bob** **Cora** **Devon** Alice Alice Alice Devon Bob
johnston-linear-matrix-algebra_TAB_002	89	—	`\begin{tabular}[t]{|l|} \hline Cora's list \\ \hline Alice \\ \hline Bob \\ \hline Cora \\ \hline Devon \\ \hline \end{tabular}`	Cora's list Alice Bob Cora Devon	Cora's list **Alice** Bob Cora Devon
johnston-linear-matrix-algebra_TAB_003	89	—	`\begin{tabular}[t]{|l|l|} \hline \multicolumn{2}{|l|}{Bob's lists:} \\ \hline Alice & Devon \\ \hline Bob & Alice \\ \hline Cora & Bob \\ \hline Devon & \\ \hline \end{tabular}`	Bob's lists: Alice | Devon Bob | Alice Cora | Bob Devon |	Bob's lists: **Alice** **Devon** Bob Alice Cora Bob Devon
johnston-linear-matrix-algebra_TAB_004	89	—	`\begin{tabular}[t]{|l|l|} \hline \multicolumn{2}{|l|}{Devon's lists:} \\ \hline Alice & Bob \\ \hline Bob & Alice \\ \hline Cora & Devon \\ \hline Devon & \\ \hline \end{tabular}`	Devon's lists: Alice | Bob Bob | Alice Cora | Devon Devon |	Devon's lists: **Alice** **Bob** Bob Alice Cora Devon Devon
johnston-linear-matrix-algebra_TAB_005	116	—	`\begin{tabular}[t]{|l|l|l|l|l|l|l|} \hline Year: & 1970 & 1980 & 1990 & 2000 & 2010 & 2020 \\ \hline Pop.: & 3.70 & 4.46 & 5.33 & 6.14 & 6.96 & 7.79 \\ \hline \end{tabular}`	Year: | 1970 | 1980 | 1990 | 2000 | 2010 | 2020 Pop.: | 3.70 | 4.46 | 5.33 | 6.14 | 6.96 | 7.79	**Year:** 1970 1980 1990 2000 2010 2020 **Pop.:** 3.70 4.46 5.33 6.14 6.96 7.79

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johnston-linear-matrix-algebra_TAB_006	188	—	`\begin{tabular}[t]{|l|l|l|l|l|l|l|l|l|l|} \hline 1 & & 1 & & . & . & . & . & . & 1 \\ \hline - & 1 & 1 & & & 1 & & & & 1 \\ \hline 1 & 1 & 1 & 1 & & & 1 & & & 1 \\ \hline & & 1 & 1 & & & & 1 & & 1 \\ \hline & & & & 1 & 1 & & & & 0 \\ \hline & 1 & & & 1 & 1 & 1 & & & 0 \\ \hline & & 1 & & & 1 & 1 & 1 & & 0 \\ \hline & & & 1 & & & 1 & 1 & 1 & 0 \\ \hline . & & & . & . & . & & 1 & 1 & 1 \\ \hline \end{tabular}`	1 | · 1 · · · · · · 1 · 1 1 · · 1 · · · 1 1 1 1 1 · · 1 · · 1 · · 1 1 · · · 1 · 1 · · · · 1 1 · · · 0 · 1 · · 1 1 1 · · 0 · · 1 · · 1 1 1 · 0 · · · 1 · · 1 1 1 0 · · · · · · · 1 1 1	
johnston-linear-matrix-algebra_TAB_007	190	—	`\begin{tabular}[t]{|l|l|l|l|} \hline + & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 2 \\ \hline 1 & 1 & 2 & 0 \\ \hline 2 & 2 & 0 & 1 \\ \hline \end{tabular}`	+ | 0 | 1 | 2 0 | 0 | 1 | 2 1 | 1 | 2 | 0 2 | 2 | 0 | 1	
johnston-linear-matrix-algebra_TAB_008	190	—	`\begin{tabular}[t]{|l|l|l|l|} \hline × & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 2 \\ \hline 2 & 0 & 2 & 1 \\ \hline \end{tabular}`	× | 0 | 1 | 2 0 | 0 | 0 | 0 1 | 0 | 1 | 2 2 | 0 | 2 | 1	
johnston-linear-matrix-algebra_TAB_009	193	—	`\begin{tabular}[t]{|l|l|l|l|l|} \hline Value & Color (c) & Shape ( \(p\) ) & Shading \((d)\) & Number ( \(n\) ) \\ \hline 0 & red & diamond & solid & one \\ \hline 1 & green & squiggle & striped & two \\ \hline 2 & purple & oval & hollow & three \\ \hline \end{tabular}`	Value | Color (c) | Shape (p) | Shading (d) | Number (n) 0 | red | diamond | solid | one 1 | green | squiggle | striped | two 2 | purple | oval | hollow | three	Value | Color (c) | Shape (p) | Shading (d) | Number (n) 0 | red | diamond | solid | one 1 | green | squiggle | striped | two 2 | purple | oval | hollow | three
johnston-linear-matrix-algebra_TAB_010	194	—	`\begin{tabular}[t]{|l|l|l|l|l|} \hline + & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 1 & 2 & 3 \\ \hline 1 & 1 & 2 & 3 & 0 \\ \hline 2 & 2 & 3 & 0 & 1 \\ \hline 3 & 3 & 0 & 1 & 2 \\ \hline \end{tabular}`	+ | 0 | 1 | 2 | 3 0 | 0 | 1 | 2 | 3 1 | 1 | 2 | 3 | 0 2 | 2 | 3 | 0 | 1 3 | 3 | 0 | 1 | 2	+ | 0 | 1 | 2 | 3 0 | 0 | 1 | 2 | 3 1 | 1 | 2 | 3 | 0 2 | 2 | 3 | 0 | 1 3 | 3 | 0 | 1 | 2

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johnston-linear-matrix-algebra_TAB_011	194	—	`\begin{tabular}[t]{|l|l|l|l|l|} \hline × & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 2 & 3 \\ \hline 2 & 0 & 2 & 0 & 2 \\ \hline 3 & 0 & 3 & 2 & 1 \\ \hline \end{tabular}`		
johnston-linear-matrix-algebra_TAB_012	194	—	`\begin{tabular}[t]{|l|l|l|l|l|l|} \hline + & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 2 & 3 & 4 & 0 \\ \hline 2 & 2 & 3 & 4 & 0 & 1 \\ \hline 3 & 3 & 4 & 0 & 1 & 2 \\ \hline 4 & 4 & 0 & 1 & 2 & 3 \\ \hline \end{tabular}`		
johnston-linear-matrix-algebra_TAB_013	194	—	`\begin{tabular}[t]{|l|l|l|l|l|l|} \hline × & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 2 & 3 & 4 \\ \hline 2 & 0 & 2 & 4 & 1 & 3 \\ \hline 3 & 0 & 3 & 1 & 4 & 2 \\ \hline 4 & 0 & 4 & 3 & 2 & 1 \\ \hline \end{tabular}`		
johnston-linear-matrix-algebra_TAB_014	206	—	`\begin{tabular}[t]{|l|} \hline \begin{tabular}[t]{l} \(\begin{array}{lllllll}z & s_{1} & s_{2} & s_{3} & x_{1} & x_{2} & x_{3}\end{array}\) \[\left[\begin{array}{c|ccc|ccc|c} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ \hline 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array}\right] \] \[\xrightarrow{ \substack{R_{1}+2 R_{3} \\ R_{2}+3 R_{3} \\ R_{4}+R_{3}}} \left[\begin{array}{c|ccc|ccc|c} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ \hline 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array}\right] . \] \\ new "leading" entry \end{tabular} \\ \hline \end{tabular}`	*(not rendered)*	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_TAB_015	209	—	\begin{tabular}[t]{|l|l|l|l|l|l|l|l|l|} \hline 1 & 0 & 0 & 0 & -1 & -0.8 & -1.5 & 0 & \\ \hline 0 & 1 & 0 & 0 & 0 & 1 & 1 & 100 & \\ \hline 0 & 0 & 1 & 0 & 1 & 0 & 1 & 80 & . \\ \hline 0 & 0 & 0 & 1 & 1 & 1 & 1 & 150 & \\ \hline \end{tabular}	1 | 0 | 0 | 0 | -1 | -0.8 | -1.5 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 100 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 80 | . 0 | 0 | 0 | 1 | 1 | 1 | 1 | 150 |	$\left[\begin{array}{c\|c\|c\|c\|c\|c\|c\|c} 1 & 0 & 0 & 0 & -1 & -0.8 & -1.5 & 0 \\ \hline 0 & 1 & 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 80 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 150 \end{array}\right].$
johnston-linear-matrix-algebra_TAB_016	210	—	\begin{tabular}[t]{|l|l|l|l|l|l|l|l|l|} \hline \multirow{2}{*}{} & \(z\) & \(s_{1}\) & \(s_{2}\) & \(s_{3}\) & \(x_{\mathrm{p}}\) & \(x_{\mathrm{c}}\) & \(x_{\mathrm{pc}}\) & \\ \hline \(\left[\begin{array}{l}1 \\ \hline 0 \\ 0 \\ 0\end{array}\right. \) & 0100 & 0 & 0 & -1 & -0.8 & -1.5 & 0 \\ \hline \(R_{1}+1.5 R_{3}\) & \(\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0\end{array}\right.\) & 0100 & 1.5 -1 1 -1 & & 0.5 & -0.8 & 0 & 120 \\ \hline \(\xrightarrow{R_{1}+0.8 R_{2}}\) & \(\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0 \end{array}\right.\) & 0.8 1 0 -1 & & 0.7 -1 1 0 & 0001 & -0.3 & 0 & 136 \\ \hline \(R_{1}+0.3 R_{4}\) & \(\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0\end{array}\right.\) & 0.5 0 1 -1 & & 0.7 -1 1 0 & 0.3 & 0 & 0 & \\ \hline \end{tabular}	*(not rendered)*	$\begin{array}{c} \begin{array}{ccccccc} z & s_1 & s_2 & s_3 & x_{\mathrm{p}} & x_{\mathrm{c}} & x_{\mathrm{pc}} \end{array} \\ \left[\begin{array}{c\|ccc\|ccc\|c} 1 & 0 & 0 & 0 & -1 & -0.8 & -1.5 & 0 \\ \hline 0 & 1 & 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 80 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 150 \end{array}\right] \\ \xrightarrow[R_4-R_3]{\substack{R_1+1.5R_3 \\ R_2-R_3}} \left[\begin{array}{c\|ccc\|ccc\|c} 1 & 0 & 1.5 & 0 & 0.5 & -0.8 & 0 & 120 \\ \hline 0 & 1 & -1 & 0 & -1 & 1 & 0 & 20 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 80 \\ 0 & 0 & -1 & 1 & 0 & 1 & 0 & 70 \end{array}\right] \\ \xrightarrow[R_4-R_2]{\substack{R_1+0.8R_2 \\ R_4-R_2}} \left[\begin{array}{c\|ccc\|ccc\|c} 1 & 0.8 & 0.7 & 0 & -0.3 & 0 & 0 & 136 \\ \hline 0 & 1 & -1 & 0 & -1 & 1 & 0 & 20 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 80 \\ 0 & -1 & 0 & 1 & 1 & 0 & 0 & 50 \end{array}\right] \\ \xrightarrow[R_3-R_4]{\substack{R_1+0.3R_4 \\ R_2+R_4}} \left[\begin{array}{c\|ccc\|ccc\|c} 1 & 0.5 & 0.7 & 0.3 & 0 & 0 & 0 & 151 \\ \hline 0 & 0 & -1 & 1 & 0 & 1 & 0 & 70 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 30 \\ 0 & -1 & 0 & 1 & 1 & 0 & 0 & 50 \end{array}\right] \end{array}$
johnston-linear-matrix-algebra_TAB_017	213	—	\begin{tabular}[t]{|l|l|l|l|l|l|l|l|l|} \hline & [& 00 & 1 & -2 & 1 & -2 & \\ \hline \((1 / 2) R_{4}\) & & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline \(R_{1}+2 R_{4}\) & 0 & 1 & 1/2 & 0 & 3/2 & -1/2 & 1 \\ \hline \(R_{2}-R_{4}\) & 0 & 0 & -1/2 & 0 & -3/2 & -1/2 & 0 \\ \hline \(R_{3}-R_{4}\) & [& 01000 & -1/2 & 1 & -1/2 & 1/2 & 1 \\ \hline \end{tabular}	| [| 00 | 1 | -2 | 1 | -2 | (1/2)R_4 | 0 | 0 | 0 | 0 | 1 | 0 $R_1 + 2R_4$ | 0 | 1 | 1/2 | 0 | 3/2 | -1/2 | 1 $R_2 - R_4$ | 0 | 0 | -1/2 | 0 | -3/2 | -1/2 | 0 $R_3 - R_4$ | [| 01000 | -1/2 | 1 | -1/2 | 1/2 | 1	$\begin{array}{c} \left[\begin{array}{c\|ccc\|ccc\|c} 1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2 \end{array}\right] \\ \xrightarrow[R_3-R_4]{\substack{(1/2)R_4 \\ R_1+2R_4 \\ R_2-R_4}} \left[\begin{array}{c\|ccc\|ccc\|c} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 1 & 0 & 1/2 & 0 & 3/2 & -1/2 & 1 \\ 0 & 0 & 1 & -1/2 & 0 & -3/2 & -1/2 & 0 \\ 0 & 0 & 0 & -1/2 & 1 & -1/2 & 1/2 & 1 \end{array}\right] \end{array}$

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johnston-linear-matrix-algebra_TAB_018	213	—	\begin{tabular}[t]{|l|l|l|l|l|l|} \hline \multirow{3}{*}{} & \(z\) & \(s_{3}\) & \(x_{1}\) & \(x_{2}\) & \\ \hline \multirow{2}{*}{} & \multicolumn{3}{|r|}{\multirow[t]{2}{*}{ \begin{tabular}[t]{l} -1 \\ 0 \\ 0 \\ -1 \\ -1 \\ -1 \end{tabular}}} & -2 & \(\left.0 \begin{array}{l}0 \\ 2 \\ -1 \\ -2\end{array}\right]\) \\ \hline & & & & & \ \ \hline \multirow[t]{2}{*}{\(R_{1}+R_{2}\)} & \(\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0\end{array}\right.\) & 0001 & 0100 & -1 & 2212 \\ \hline & & & & & \\ \hline \end{tabular}	*(not rendered)*	$\begin{array}{c} \begin{array}{cccccc} z & s_1 & s_2 & s_3 & x_1 & x_2 \end{array} \\ \left[\begin{array}{c\|ccc\|cc\|c} 1 & 0 & 0 & 0 & -1 & -2 & 0 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -2 \end{array}\right] \\ \xrightarrow[R_4+2R_2]{\substack{R_1+R_2 \\ R_3+R_2}} \left[\begin{array}{c\|ccc\|cc\|c} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\right] \end{array}$
johnston-linear-matrix-algebra_TAB_019	214	—	\begin{tabular}[t]{|l|l|} \hline \multirow[b]{2}{*}{\(\xrightarrow{\substack{(1 / 3) R_{4} \\ R_{1}+R_{4} \ \ R_{2}-R_{4}}}\)} & \[\left[\begin{array}{c|ccc|cc|c} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\right] \] \\ \hline & \(\left[\begin{array}{c|ccc|cc|c}1 & 5 / 3 & 0 & 1 / 3 & 0 & 0 & 8 / 3 \\ \hline 0 & 1 / 3 & 0 & -1 / 3 & 1 & 0 & 4 / 3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 / 3 & 0 & 1 / 3 & 0 & 1 & 2 / 3 \end{array}\right]\) \\ \hline \end{tabular}	*(not rendered)*	$\xrightarrow[\substack{R_1+R_4 \\ R_2-R_4}]{(1/3)R_4} \left[\begin{array}{c\|ccc\|cc\|c} 1 & 5/3 & 0 & 1/3 & 0 & 0 & 8/3 \\ \hline 0 & 1/3 & 0 & -1/3 & 1 & 0 & 4/3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2/3 & 0 & 1/3 & 0 & 1 & 2/3 \end{array}\right]$
johnston-linear-matrix-algebra_TAB_020	219	—	\begin{tabular}[t]{|l|l|l|} \hline & Primal & Dual \\ \hline maximize: & \(x_{1}+2 x_{2}\) & minimize: \(\quad y\) \\ \hline subject to: & \(-x_{1}+x_{2} \leq 1\) & subject to: \(\quad-y \geq 1\) \\ \hline & \(x_{1}, \quad x_{2} \geq 0\) & \(y \geq 2\) \\ \hline & & \(y \geq 0\) \\ \hline \end{tabular}	\begin{tabular}[t]{|l|l|l|} \hline & Primal & Dual \\ \hline maximize: & $x_1 + 2x_2$ & minimize: y \\ \hline subject to: & $-x_1 + x_2 \leq 1$ & subject to: $-y \geq 1$ \\ \hline & $x_1, \quad x_2 \geq 0$ & $y \geq 2$ \\ \hline & & $y \geq 0$ \\ \hline \end{tabular}	**Primal** $\begin{array}{ll} \text{maximize:} & x_1 + 2x_2 \\ \text{subject to:} & -x_1 + x_2 \leq 1 \\ & x_1, \quad x_2 \geq 0 \end{array}$ **Dual** $\begin{array}{ll} \text{minimize:} & y \\ \text{subject to:} & -y \geq 1 \\ & y \geq 2 \\ & y \geq 0 \end{array}$
johnston-linear-matrix-algebra_TAB_021	219	—	\begin{tabular}[t]{|l|l|l|} \hline & Primal & Dual \ \ \hline maximize: & \(x_{1}+2 x_{2}\) & minimize: \(-y_{1}-y_{2}\) \\ \hline subject to: & \(x_{1}-x_{2} \leq-1\) & subject to: \(\quad y_{1}-y_{2} \geq 1\) \\ \hline & \(-x_{1}+x_{2} \leq-1\) & \(-y_{1}+y_{2} \geq 2 \) \\ \hline & \(x_{1}, \quad x_{2} \geq 0\) & \(y_{1}, \quad y_{2} \geq 0\) \\ \hline \end{tabular}	\begin{tabular}[t]{|l|l|l|} \hline & Primal & Dual \\ \hline maximize: & $x_1 + 2x_2$ & minimize: $-y_1 - y_2$ \\ \hline subject to: & $x_1 - x_2 \leq -1$ & subject to: $y_1 - y_2 \geq 1$ \\ \hline & $-x_1 + x_2 \leq -1$ & $-y_1 + y_2 \geq 2$ \\ \hline & $x_1, \quad x_2 \geq 0$ & $y_1, \quad y_2 \geq 0$ \\ \hline \end{tabular}	**Primal** $\begin{array}{ll} \text{maximize:} & x_1 + 2x_2 \\ \text{subject to:} & x_1 - x_2 \leq -1 \\ & -x_1 + x_2 \leq -1 \\ & x_1, \quad x_2 \geq 0 \end{array}$ **Dual** $\begin{array}{ll} \text{minimize:} & -y_1 - y_2 \\ \text{subject to:} & y_1 - y_2 \geq 1 \\ & -y_1 + y_2 \geq 2 \\ & y_1, \quad y_2 \geq 0 \end{array}$

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johnston-linear-matrix-algebra_TAB_022	219	—	`\begin{tabular}[t]{|l|l|l|l|l|} \hline \multicolumn{2}{|c|}{\multirow{2}{*}{}} & \multicolumn{3}{|c|}{Primal problem} \\ \hline & & Infeasible & Solvable & Unbounded \\ \hline \multirow[b]{3}{*}{Dual} & Infeasible & ✓ & . & ✓ \\ \hline & Solvable & . & ✓ & . \\ \hline & Unbounded & ✓ & . & . \\ \hline \end{tabular}`	*(not rendered)*	
johnston-linear-matrix-algebra_TAB_023	221	—	`\begin{tabular}[t]{|l|l|l|l|} \hline \multicolumn{2}{|c|}{Primal problem (maximize)} & \multicolumn{2}{|c|}{Dual problem (minimize)} \\ \hline \(i\)-th constraint: \[\begin{aligned} & \geq \\ & \leq \\ & \leq \end{aligned} \] & & & \(i\)-th variable: \[\begin{aligned} & \leq 0 \\ & \text { unc } \\ & \geq 0 \end{aligned} \] \\ \hline & \(j\)-th variable: \[\begin{aligned} & \geq 0 \\ & \text { unconstrained } \\ & \leq 0 \end{aligned} \] & & \(j\)-th constraint: \[\begin{aligned} & \geq \\ & = \\ & \leq \end{aligned} \] & \\ \hline \end{tabular}`	*(not rendered)*	
johnston-linear-matrix-algebra_TAB_024	244	—	`\begin{tabular}[t]{|l|l|l|} \hline Algorithm & Resulting form & Matrix decomposition \\ \hline Gaussian elimination & row echelon form & PLU decomposition \\ \hline Gauss-Jordan elim. & reduced REF & RREF decomposition \\ \hline \end{tabular}`	Algorithm | Resulting form | Matrix decomposition Gaussian elimination | row echelon form | PLU decomposition Gauss-Jordan elim. | reduced REF | RREF decomposition	
johnston-linear-matrix-algebra_TAB_025	366	—	`\begin{tabular}[t]{|l|l|l|} \hline \(\sigma\) & \(\operatorname{sgn}(\sigma)\) & contribution to \(\operatorname{det}(A)\) \\ \hline (12 3) & 1 & \(a_{1,1} a_{2,2} a_{3,3}\) \\ \hline (1 3 2) & -1 & \(-a_{1,1} a_{3,2} a_{2,3}\) \\ \hline (213) & -1 & \(-a_{2,1} a_{1,2} a_{3,3}\) \\ \hline (2 3 1) & 1 & \(a_{2,1} a_{3,2} a_{1,3}\) \\ \hline (312) & 1 & \(a_{3,1} a_{1,2} a_{2,3}\) \\ \hline (321) & -1 & \(-a_{3,1} a_{2,2} a_{1,3}\) \\ \hline \end{tabular}`	σ | $\operatorname{sgn}(\sigma)$ | contribution to $\operatorname{det}(A)$ (12 3) | 1 | $a_{1,1} a_{2,2} a_{3,3}$ (1 3 2) | -1 | $-a_{1,1} a_{3,2} a_{2,3}$ (213) | -1 | $-a_{2,1} a_{1,2} a_{3,3}$ (2 3 1) | 1 | $a_{2,1} a_{3,2} a_{1,3}$ (312) | 1 | $a_{3,1} a_{1,2} a_{2,3}$ (321) | -1 | $-a_{3,1} a_{2,2} a_{1,3}$	

Image

		Primal problem		
		Infeasible	Solvable	Unbounded
Dual	Infeasible	✓	·	✓
	Solvable	·	✓	·
	Unbounded	✓	·	·

Primal problem (maximize)		Dual problem (minimize)	
i-th constraint:	$\geq$	i-th variable:	≤ 0
	$=$		unconstrained
	$\leq$		≥ 0
j-th variable:	≥ 0	j-th constraint:	$\geq$
	unconstrained		$=$
	≤ 0		$\leq$

Algorithm	Resulting form	Matrix decomposition
Gaussian elimination	row echelon form	PLU decomposition
Gauss–Jordan elim.	reduced REF	RREF decomposition

σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$
(1 2 3)	1	$a_{1,1} a_{2,2} a_{3,3}$
(1 3 2)	-1	$-a_{1,1} a_{3,2} a_{2,3}$
(2 1 3)	-1	$-a_{2,1} a_{1,2} a_{3,3}$
(2 3 1)	1	$a_{2,1} a_{3,2} a_{1,3}$
(3 1 2)	1	$a_{3,1} a_{1,2} a_{2,3}$
(3 2 1)	-1	$-a_{3,1} a_{2,2} a_{1,3}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_TAB_026	372	—	\begin{tabular}[t]{|l|l|l|l|} \hline \(k\) & \(A \mathbf{v}_{k-1}\) & \(\left|A \mathbf{v}_{k-1}\right |\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left|A \mathbf{v}_{k-1}\right|\) \\ \hline 0 & - & - & \(\mathbf{v}_{0}=\mathbf{e}_{1}=(1,0)\) \\ \hline 1 & (1.00, 5.00) & 5.10 & (0.20, 0.98) \\ \hline 2 & (2.16,4.90) & 5.36 & (0.40, 0.92) \\ \hline 3 & (2.23, 5.68) & 6.10 & (0.37, 0.93) \\ \hline 4 & (2.23, 5.55) & 5.98 & (0.37, 0.93) \\ \hline 5 & (2.23, 5.57) & 6.00 & (0.37, 0.93) \\ \hline \end{tabular}	k | $A\mathbf{v}_{k-1}$ | $\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ 0 | - | - | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$ 1 | (1.00, 5.00) | 5.10 | (0.20, 0.98) 2 | (2.16,4.90) | 5.36 | (0.40, 0.92) 3 | (2.23, 5.68) | 6.10 | (0.37, 0.93) 4 | (2.23, 5.55) | 5.98 | (0.37, 0.93) 5 | (2.23, 5.57) | 6.00 | (0.37, 0.93)	k | $A\mathbf{v}_{k-1}$ | $\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ 0 | – | – | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$ 1 | $(1.00, 5.00)$ | 5.10 | $(0.20, 0.98)$ 2 | $(2.16, 4.90)$ | 5.36 | $(0.40, 0.92)$ 3 | $(2.23, 5.68)$ | 6.10 | $(0.37, 0.93)$ 4 | $(2.23, 5.55)$ | 5.98 | $(0.37, 0.93)$ 5 | $(2.23, 5.57)$ | 6.00 | $(0.37, 0.93)$
johnston-linear-matrix-algebra_TAB_027	373	—	\begin{tabular}[t]{|l|l|l|l|l|l|l|l|} \hline \(k \) & \multicolumn{6}{|c|}{\(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left|A \mathbf{v}_{k-1}\right|\)} & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, & 0.00, & 0.00, & 0.00, & 0.00, & 0.00) & 0.00 \\ \hline 1 & (0.00, -0.50, & 0.50, & 0.00, & 0.50, & 0.00, & 0.50) & -2.00 \\ \hline 2 & (0.59, & 0.44, & 0.29, & 0.44, & 0.29, & 0.00) & -4.26 \\ \hline 3 & (-0.26, & 0.51, & 0.32, & 0.51, & 0.32, & 0.00) & -4.83 \\ \hline 4 & (0.30, & 0.49, & 0.37, & 0.49, & 0.37, & 0.08) & -4.91 \\ \hline 5 & (-0.26, & 0.49, & 0.37, & 0.49, & 0.37, & 0.09) & -4.91 \\ \hline 6 & (0.27, & 0.49, & 0.38, & 0.49, & 0.38, & 0.10) & -4.92 \\ \hline 7 & (-0.26, & 0.49, & 0.38, & 0.49, & 0.38, & 0.10) & -4.92 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, | 0.00, | 0.00, | 0.00, | 0.00, | 0.00) | 0.00 1 | (0.00, -0.50, | 0.50, | 0.00, | 0.50, | 0.00, | 0.50) | -2.00 2 | (0.59, | 0.44, | 0.29, | 0.44, | 0.29, | 0.00) | -4.26 3 | (-0.26, | 0.51, | 0.32, | 0.51, | 0.32, | 0.00) | -4.83 4 | (0.30, | 0.49, | 0.37, | 0.49, | 0.37, | 0.08) | -4.91 5 | (-0.26, | 0.49, | 0.37, | 0.49, | 0.37, | 0.09) | -4.91 6 | (0.27, | 0.49, | 0.38, | 0.49, | 0.38, | 0.10) | -4.92 7 | (-0.26, | 0.49, | 0.38, | 0.49, | 0.38, | 0.10) | -4.92	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00)$ | 0.00 1 | $(0.00, -0.50, 0.50, 0.00, 0.50, 0.00, 0.50)$ | -2.00 2 | $(0.59, 0.29, -0.44, -0.29, -0.44, 0.29, 0.00)$ | -4.26 3 | $(-0.26, -0.45, 0.51, 0.32, 0.51, -0.32, 0.00)$ | -4.83 4 | $(0.30, 0.39, -0.49, -0.37, -0.49, 0.37, 0.08)$ | -4.91 5 | $(-0.26, -0.41, 0.49, 0.37, 0.49, -0.37, -0.09)$ | -4.91 6 | $(0.27, 0.40, -0.49, -0.38, -0.49, 0.38, 0.10)$ | -4.92 7 | $(-0.26, -0.41, 0.49, 0.38, 0.49, -0.38, -0.10)$ | -4.92

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_TAB_028	374	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=C \mathbf{v}_{k-1} /\left\\|C \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} C \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00, 0.00, 0.00, 0.00) & 0.00 \\ \hline 1 & (0.00, 0.00, 0.00, 0.00, 0.00, 1.00) & 2.00 \\ \hline 2 & (0.00, 0.00, 0.00, 0.00, 0.45, 0.89) & 1.20 \\ \hline 3 & (0.00, 0.00, 0.00, 0.33, 0.67, 0.67) & 1.33 \\ \hline & **[U+22EE]** & \\ \hline 26 & (0.06, 0.10, 0.17, 0.28, 0.48, 0.81) & 1.69 \\ \hline 27 & (0.06, 0.10, 0.17, 0.28, 0.48, 0.81) & 1.68 \\ \hline \end{tabular}	k | $\mathbf{v}_k = C\mathbf{v}_{k-1}/\lVert C\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T C\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00, 0.00, 0.00, 0.00) | 0.00 1 | (0.00, 0.00, 0.00, 0.00, 0.00, 1.00) | 2.00 2 | (0.00, 0.00, 0.00, 0.00, 0.45, 0.89) | 1.20 3 | (0.00, 0.00, 0.00, 0.33, 0.67, 0.67) | 1.33 **[U+22EE]** 26 | (0.06, 0.10, 0.17, 0.28, 0.48, 0.81) | 1.69 27 | (0.06, 0.10, 0.17, 0.28, 0.48, 0.81) | 1.68	k | $\mathbf{v}_k = C\mathbf{v}_{k-1}/\lVert C\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T C\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00, 0.00, 0.00, 0.00)$ | 0.00 1 | $(0.00, 0.00, 0.00, 0.00, 0.00, 1.00)$ | 2.00 2 | $(0.00, 0.00, 0.00, 0.00, 0.45, 0.89)$ | 1.20 3 | $(0.00, 0.00, 0.00, 0.33, 0.67, 0.67)$ | 1.33 $\vdots$ 26 | $(0.06, 0.10, 0.17, 0.28, 0.48, 0.81)$ | 1.69 27 | $(0.06, 0.10, 0.17, 0.28, 0.48, 0.81)$ | 1.68
johnston-linear-matrix-algebra_TAB_029	375	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00) & 1.00 \\ \hline 1 & (1.00, 0.00) & 1.00 \\ \hline 2 & (1.00, 0.00) & 1.00 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00) | 1.00 1 | (1.00, 0.00) | 1.00 2 | (1.00, 0.00) | 1.00	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(1.00, 0.00)$ | 1.00 1 | $(1.00, 0.00)$ | 1.00 2 | $(1.00, 0.00)$ | 1.00
johnston-linear-matrix-algebra_TAB_030	375	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (0.80, 0.60) & 2.68 \\ \hline 1 & (0.74, 0.67) & 2.89 \\ \hline 2 & (0.72, 0.69) & 2.96 \\ \hline 3 & (0.71, 0.70) & 2.99 \\ \hline 4 & (0.71, 0.71) & 3.00 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (0.80, 0.60) | 2.68 1 | (0.74, 0.67) | 2.89 2 | (0.72, 0.69) | 2.96 3 | (0.71, 0.70) | 2.99 4 | (0.71, 0.71) | 3.00	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(0.80, 0.60)$ | 2.68 1 | $(0.74, 0.67)$ | 2.89 2 | $(0.72, 0.69)$ | 2.96 3 | $(0.71, 0.70)$ | 2.99 4 | $(0.71, 0.71)$ | 3.00
johnston-linear-matrix-algebra_TAB_031	376	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00) & 1.00 \\ \hline 1 & (0.32, 0.95) & 0.40 \\ \hline 2 & (1.00, 0.00) & 1.00 \\ \hline 3 & (0.32, 0.95) & 0.40 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00) | 1.00 1 | (0.32, 0.95) | 0.40 2 | (1.00, 0.00) | 1.00 3 | (0.32, 0.95) | 0.40	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(1.00, 0.00)$ | 1.00 1 | $(0.32, 0.95)$ | 0.40 2 | $(1.00, 0.00)$ | 1.00 3 | $(0.32, 0.95)$ | 0.40

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johnston-linear-matrix-algebra_TAB_032	376	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (0.80, 0.60) & 2.20 \\ \hline 1 & (0.61, 0.79) & 1.69 \\ \hline 2 & (0.80, 0.60) & 2.20 \\ \hline 3 & (0.61, 0.79) & 1.69 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (0.80, 0.60) | 2.20 1 | (0.61, 0.79) | 1.69 2 | (0.80, 0.60) | 2.20 3 | (0.61, 0.79) | 1.69	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(0.80, 0.60)$ | 2.20 1 | $(0.61, 0.79)$ | 1.69 2 | $(0.80, 0.60)$ | 2.20 3 | $(0.61, 0.79)$ | 1.69
johnston-linear-matrix-algebra_TAB_033	378	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00) & 8.00 \\ \hline 1 & (0.99, -0.12, 0.00) & 8.15 \\ \hline 2 & (0.99, -0.15, -0.07) & 8.19 \\ \hline 3 & (0.98, -0.20, -0.04) & 8.21 \\ \hline 4 & (0.98, -0.19, -0.10) & 8.21 \\ \hline 5 & (0.97, -0.22, -0.05) & 8.21 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00) | 8.00 1 | (0.99, -0.12, 0.00) | 8.15 2 | (0.99, -0.15, -0.07) | 8.19 3 | (0.98, -0.20, -0.04) | 8.21 4 | (0.98, -0.19, -0.10) | 8.21 5 | (0.97, -0.22, -0.05) | 8.21	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00)$ | 8.00 1 | $(0.99, -0.12, 0.00)$ | 8.15 2 | $(0.99, -0.15, -0.07)$ | 8.19 3 | $(0.98, -0.20, -0.04)$ | 8.21 4 | $(0.98, -0.19, -0.10)$ | 8.21 5 | $(0.97, -0.22, -0.05)$ | 8.21
johnston-linear-matrix-algebra_TAB_034	379	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 318 & (0.05, 0.43, -0.90) & -8.40 \\ \hline 319 & (0.00, -0.44, 0.90) & -8.40 \\ \hline 320 & (0.05, 0.43, -0.90) & -8.41 \\ \hline 321 & (0.00, -0.44, 0.90) & -8.41 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 318 | (0.05, 0.43, -0.90) | -8.40 319 | (0.00, -0.44, 0.90) | -8.40 320 | (0.05, 0.43, -0.90) | -8.41 321 | (0.00, -0.44, 0.90) | -8.41	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 318 | $(0.05, 0.43, -0.90)$ | -8.40 319 | $(0.00, -0.44, 0.90)$ | -8.40 320 | $(0.05, 0.43, -0.90)$ | -8.41 321 | $(0.00, -0.44, 0.90)$ | -8.41
johnston-linear-matrix-algebra_TAB_035	380	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00) & 1.00 \\ \hline 1 & (0.20, 0.78, 0.59) & 7.85 \\ \hline 2 & (0.42, 0.84, 0.33) & 8.49 \\ \hline 3 & (0.36, 0.85, 0.39) & 8.53 \\ \hline 4 & (0.38, 0.85, 0.37) & 8.53 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00) | 1.00 1 | (0.20, 0.78, 0.59) | 7.85 2 | (0.42, 0.84, 0.33) | 8.49 3 | (0.36, 0.85, 0.39) | 8.53 4 | (0.38, 0.85, 0.37) | 8.53	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00)$ | 1.00 1 | $(0.20, 0.78, 0.59)$ | 7.85 2 | $(0.42, 0.84, 0.33)$ | 8.49 3 | $(0.36, 0.85, 0.39)$ | 8.53 4 | $(0.38, 0.85, 0.37)$ | 8.53

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johnston-linear-matrix-algebra_TAB_036	380	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=B \mathbf{v}_{k-1} /\left\\|B \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} B \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00) & -0.21 \\ \hline 1 & (-0.10, 0.57, 0.81) & -0.52 \\ \hline 2 & (0.81, 0.22, -0.54) & -1.81 \\ \hline 3 & (-0.68, 0.04, 0.73) & -1.99 \\ \hline 4 & (0.72, 0.03, -0.69) & -2.00 \\ \hline \end{tabular}	k | $\mathbf{v}_k = B\mathbf{v}_{k-1}/\lVert B\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T B\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00) | -0.21 1 | (-0.10, 0.57, 0.81) | -0.52 2 | (0.81, 0.22, -0.54) | -1.81 3 | (-0.68, 0.04, 0.73) | -1.99 4 | (0.72, 0.03, -0.69) | -2.00	k | $\mathbf{v}_k = B\mathbf{v}_{k-1}/\lVert B\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T B\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00)$ | -0.21 1 | $(-0.10, 0.57, 0.81)$ | -0.52 2 | $(0.81, 0.22, -0.54)$ | -1.81 3 | $(-0.68, 0.04, 0.73)$ | -1.99 4 | $(0.72, 0.03, -0.69)$ | -2.00
johnston-linear-matrix-algebra_TAB_037	381	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=C \mathbf{v}_{k-1} /\left\\|C \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} C \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00) & 0.82 \\ \hline 1 & (0.47, 0.75, 0.46) & 0.47 \\ \hline 2 & (0.51, 0.75, 0.42) & 0.47 \\ \hline 3 & (0.51, 0.75, 0.42) & 0.47 \\ \hline \end{tabular}	k | $\mathbf{v}_k = C\mathbf{v}_{k-1}/\lVert C\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T C\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00) | 0.82 1 | (0.47, 0.75, 0.46) | 0.47 2 | (0.51, 0.75, 0.42) | 0.47 3 | (0.51, 0.75, 0.42) | 0.47	k | $\mathbf{v}_k = C\mathbf{v}_{k-1}/\lVert C\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T C\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00)$ | 0.82 1 | $(0.47, 0.75, 0.46)$ | 0.47 2 | $(0.51, 0.75, 0.42)$ | 0.47 3 | $(0.51, 0.75, 0.42)$ | 0.47
johnston-linear-matrix-algebra_TAB_038	381	—	\begin{tabular}[t]{|l|l|l|l|l|} \hline \multicolumn{2}{|c|}{Positive} & & \multicolumn{2}{|c|}{Not positive} \\ \hline \(\left[\begin{array}{ll}3 & 1 \\ 2 & 7\end{array}\right]\) & \(\left[\begin{array}{lll}2 & 4 & 1 \\ 8 & 1 & 3\end{array} \right]\) & & \(\left[\begin{array}{cc}1 & -1 \\ -2 & 3\end{array}\right]\) & \(\left[\begin{array}{lll}4 & 3 & 2 \\ 1 & 0 & 1\end{array}\right]\). \\ \hline \end{tabular}	Positive | | Not positive $\begin{bmatrix}3 & 1\\ 2 & 7\end{bmatrix}$ | $\begin{bmatrix}2 & 4 & 1\\ 8 & 1 & 3\end{bmatrix}$ | | $\begin{bmatrix}1 & -1\\ -2 & 3\end{bmatrix}$ | $\begin{bmatrix}4 & 3 & 2\\ 1 & 0 & 1\end{bmatrix}$.	**Positive** | **Not positive** $\begin{bmatrix}3 & 1\\ 2 & 7\end{bmatrix}$ $\begin{bmatrix}2 & 4 & 1\\ 8 & 1 & 3\end{bmatrix}$ | $\begin{bmatrix}1 & -1\\ -2 & 3\end{bmatrix}$ $\begin{bmatrix}4 & 3 & 2\\ 1 & 0 & 1\end{bmatrix}$.
johnston-linear-matrix-algebra_TAB_039	385	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00) & 2.00 \\ \hline 1 & (0.44, 0.87, 0.22) & 4.48 \\ \hline 2 & (0.52, 0.60, 0.60) & 6.96 \\ \hline 3 & (0.58, 0.56, 0.59) & 7.06 \\ \hline 4 & (0.58, 0.58, 0.58) & 7.00 \\ \hline 5 & (0.58, 0.58, 0.58) & 7.00 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00) | 2.00 1 | (0.44, 0.87, 0.22) | 4.48 2 | (0.52, 0.60, 0.60) | 6.96 3 | (0.58, 0.56, 0.59) | 7.06 4 | (0.58, 0.58, 0.58) | 7.00 5 | (0.58, 0.58, 0.58) | 7.00	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\lVert A\mathbf{v}_{k-1}\rVert$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | $(1.00, 0.00, 0.00)$ | 2.00 1 | $(0.44, 0.87, 0.22)$ | 4.48 2 | $(0.52, 0.60, 0.60)$ | 6.96 3 | $(0.58, 0.56, 0.59)$ | 7.06 4 | $(0.58, 0.58, 0.58)$ | 7.00 5 | $(0.58, 0.58, 0.58)$ | 7.00

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johnston-linear-matrix-algebra_TAB_040	388	—	\begin{tabular}[t]{|l|l|l|} \hline \(k\) & \(\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\\|A \mathbf{v}_{k-1}\right\\|\) & \(\mathbf{v}_{k}^{T} A \mathbf{v}_{k}\) \\ \hline 0 & (1.00, 0.00, 0.00, 0.00, 0.00) & 0.00 \\ \hline 1 & (0.00, 0.71, 0.71, 0.00, 0.00) & 0.00 \\ \hline 2 & (0.94, 0.00, 0.00, 0.24, 0.24) & 0.21 \\ \hline 3 & (0.23, 0.68, 0.68, 0.08, 0.15) & 0.49 \\ \hline & **[U+22EE]** & \\ \hline 16 & (0.74, 0.41, 0.41, 0.21, 0.25) & 0.99 \\ \hline 17 & (0.72, 0.44, 0.44, 0.20, 0.24) & 1.00 \\ \hline 18 & (0.74, 0.42, 0.42, 0.21, 0.24) & 1.00 \\ \hline \end{tabular}	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00, 0.00, 0.00) | 0.00 1 | (0.00, 0.71, 0.71, 0.00, 0.00) | 0.00 2 | (0.94, 0.00, 0.00, 0.24, 0.24) | 0.21 3 | (0.23, 0.68, 0.68, 0.08, 0.15) | 0.49 | **[U+22EE]** | 16 | (0.74, 0.41, 0.41, 0.21, 0.25) | 0.99 17 | (0.72, 0.44, 0.44, 0.20, 0.24) | 1.00 18 | (0.74, 0.42, 0.42, 0.21, 0.24) | 1.00	k | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\left\\|A\mathbf{v}_{k-1}\right\\|$ | $\mathbf{v}_k^T A\mathbf{v}_k$ 0 | (1.00, 0.00, 0.00, 0.00, 0.00) | 0.00 1 | (0.00, 0.71, 0.71, 0.00, 0.00) | 0.00 2 | (0.94, 0.00, 0.00, 0.24, 0.24) | 0.21 3 | (0.23, 0.68, 0.68, 0.08, 0.15) | 0.49 ⋮ 16 | (0.74, 0.41, 0.41, 0.21, 0.25) | 0.99 17 | (0.72, 0.44, 0.44, 0.20, 0.24) | 1.00 18 | (0.74, 0.42, 0.42, 0.21, 0.24) | 1.00
johnston-linear-matrix-algebra_TAB_041	410	—	\begin{tabular}[t]{|l|l|l|} \hline \(n\) & \(F_{n}\) & \(F_{n} / F_{n-1}\) \\ \hline 0 & 0 & - \\ \hline 1 & 1 & - \\ \hline 2 & 1 & 1.0000 \\ \hline 3 & 2 & 2.0000 \\ \hline 4 & 3 & 1.5000 \\ \hline 5 & 5 & 1.6667 \\ \hline \end{tabular}	n | F_n | F_n/F_{n-1} 0 | 0 | - 1 | 1 | - 2 | 1 | 1.0000 3 | 2 | 2.0000 4 | 3 | 1.5000 5 | 5 | 1.6667	n | F_n | F_n/F_{n-1} 0 | 0 | – 1 | 1 | – 2 | 1 | 1.0000 3 | 2 | 2.0000 4 | 3 | 1.5000 5 | 5 | 1.6667
johnston-linear-matrix-algebra_TAB_042	410	—	\begin{tabular}[t]{|l|l|l|} \hline \(n\) & \(F_{n}\) & \(F_{n} / F_{n-1}\) \\ \hline 6 & 8 & 1.6000 \\ \hline 7 & 13 & 1.6250 \\ \hline 8 & 21 & 1.6154 \\ \hline 9 & 34 & 1.6190 \\ \hline 10 & 55 & 1.6176 \\ \hline 11 & 89 & 1.6182 \\ \hline \end{tabular}	n | F_n | F_n/F_{n-1} 6 | 8 | 1.6000 7 | 13 | 1.6250 8 | 21 | 1.6154 9 | 34 | 1.6190 10 | 55 | 1.6176 11 | 89 | 1.6182	n | F_n | F_n/F_{n-1} 6 | 8 | 1.6000 7 | 13 | 1.6250 8 | 21 | 1.6154 9 | 34 | 1.6190 10 | 55 | 1.6176 11 | 89 | 1.6182

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johnston-linear-matrix-algebra_TAB_043	410	—	`\begin{tabular}[t]{|l|l|l|} \hline \(n\) & \(F_{n}\) & \(F_{n} / F_{n-1}\) \\ \hline 12 & 144 & 1.6180 \\ \hline 13 & 233 & 1.6181 \\ \hline 14 & 377 & 1.6180 \\ \hline 15 & 610 & 1.6180 \\ \hline 16 & 987 & 1.6180 \\ \hline 17 & 1597 & 1.6180 \\ \hline \end{tabular}`	n | F_n | F_n/F_{n-1} 12 | 144 | 1.6180 13 | 233 | 1.6181 14 | 377 | 1.6180 15 | 610 | 1.6180 16 | 987 | 1.6180 17 | 1597 | 1.6180	n | F_n | F_n/F_{n-1} 12 | 144 | 1.6180 13 | 233 | 1.6181 14 | 377 | 1.6180 15 | 610 | 1.6180 16 | 987 | 1.6180 17 | 1597 | 1.6180